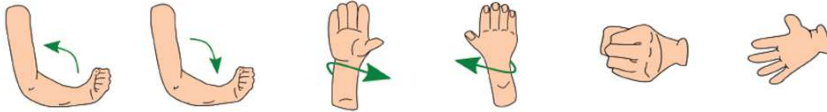


HUMAN MACHINE INTERFACES

PROJECT

Upper limb movements can be decoded from the time-domain of low-frequency EEG



HMI PROJECT CALENDAR

27/02	S3. Kick-off. Preprocessing
27/03	S7. Motor related cortical potential (MRCP).
24/04	S10. Sensorimotor rhythms. Event-related desynchronization/synchronization (ERD/ERS)
08/06	Delivery date (5-page paper)

DATABASE

- 15 healthy subjects (age: 27 ± 5 yo): 9 ♀ 6 ♂
- Motor execution / imagery task
- 7 different movements:
 - ✓ Elbow flexion / extension
 - ✓ Forearm supination / pronation
 - ✓ Hand open / close
 - ✓ Rest
- 60 repetitions / trials of each movement (~1h duration, divided into 10 runs of ~6min each)
- In each run: 6 repetitions of each movement (42 trials per run)

DATABASE - SIGNALS

▪ EEG signals

- ✓ 61 channels (frontal, central, parietal, and temporal)
- ✓ Active electrodes
- ✓ Referenced to right mastoid

▪ EOG signals

- ✓ 3 channels (left, middle and right positions)
- ✓ Referenced to right mastoid

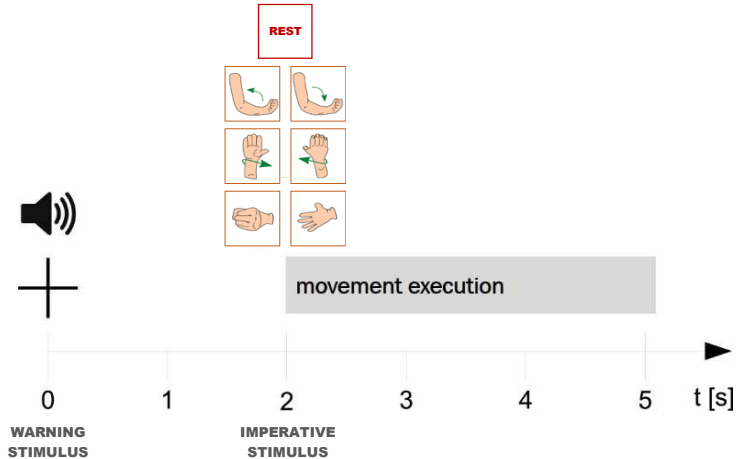
▪ Position sensors

- ✓ Exoskeleton: 13 signals (arm joint angles)
- ✓ Data glove sensors: 19 signals (finger positions)

▪ Sample frequency: 512 Hz

▪ Band-pass filtered: 0,01 – 200 Hz (Notch 50Hz)

TRIAL SEQUENCE



MAT FILES

- 1 file per run (10 runs for each subject) ME_SXX_rYY.mat
- Structured array EEG with 3 fields
 - ✓ data: matrix variable with size #channels x #samples
 - ✓ events: matrix with 3 columns
 - 1) Event codes (related to each of the 7 movement classes)
 - 2) Latency (exact sample where the imperative stimulus was presented)
 - 3) Movement onset (exact sample where the movement onset was detected)
 - ✓ chanlocs: structure array with different fields corresponding to channels labels and locations

BCI PAPER

- The proposed paper should focus on an analysis conducting using the database.
- Initiative and originality will be valued. Some potential topics include
 - 1) Handling with bad channels and bad trials: comparison between z-score and Tukey methodologies.
 - 2) Comparison of motor-evoked potentials (MRCP and ERD/ERS) based on different EEG reference strategies (CAR, Laplacian, Rest).
 - 3) Influence of ICA algorithms (Infomax, FastICA, SOBI) on motor-evoked potentials, focusing of different artifact classes).
 - 4) Effect of ICLabel thresholds on motor-evoked potentials (MRCP and ERD/ERS).
 - 5) Comparison between movement classes based on different feature sets.
 - 6) Calculation of the Lateralized Readiness Potential (LRP) and its comparison with MRCP and ERD/ERS.
 - 7) Subject-specific customization of mu and beta band ranges (reactive band identification).
 - 8) Movement-offset potentials (ERS) and potentials associated with returning to the initial position.